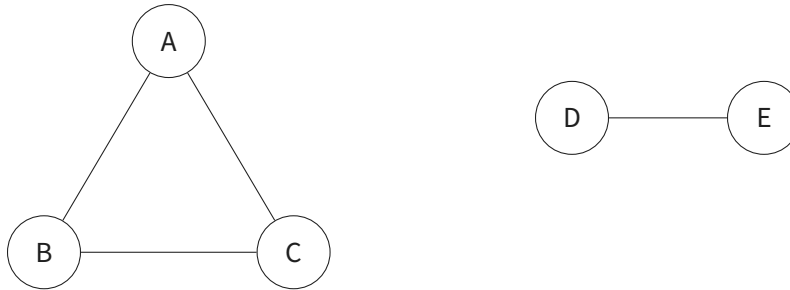


A theorem for any network

Name: _____

Date: _____

Start every lamp OFF. One move flips the two ends of one drawn edge. A target is a chosen set of lamps to leave ON.



1. Is even enough?

Can you light only A and D? Only A, C, D, E? Give a move list for one and an obstruction for the other.

My results and reasons:

2. Repair the conjecture.

What must be true of the number of target lamps in **each connected piece** of a network? Prove necessity. Then prove sufficiency by pairing target lamps and following paths. If paths overlap, what happens to a shared edge?

An if-and-only-if rule for every finite network:

3. Find the cheapest repair on a tree.

Draw a connected network with no closed loops (a **tree**), using at least six dots. Choose an even number of target lamps. Removing an edge splits the tree in two. Prove that an edge is pressed an **odd number of times** exactly when one side has an odd number of targets. Why is the solution using each edge at most once **unique and shortest**?

A labeled example and an argument:

Try it with objects. Explain by pointing, talking, or drawing.